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Experimental Study of CO₂-Brine Two-Phase Flow in Sandstone Rock

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Abstract

The experimental study of carbon dioxide injection is primarily associated with CO₂ geological storage in deep aquifers. The application of this technology requires detailed laboratory investigation of CO₂ interaction mechanisms with formation rocks and fluids under reservoir conditions. The effectiveness of field-scale operations depends heavily on accurate core analysis that simulates simultaneous reservoir fluid flow. This study addresses the physicochemical behavior of CO₂ and brine flow in siderite-containing sandstone rocks using laboratory techniques. Relative permeability measurements for brine and CO₂ were conducted using X-ray radiography and low-field nuclear magnetic resonance (NMR) relaxometry. X-ray radiography enabled quantification of end effects and accurate relative permeability measurement in drainage and imbibition modes, while NMR relaxometry provided detailed fluid distribution across pores of varying sizes. Furthermore, this work investigates CO₂ diffusion in porous media, combining experimental measurements under ambient conditions with pore-network modeling under reservoir conditions. The integrated use of these methods offers a robust characterization approach for two-phase flow in porous media, enhancing the reliability of reservoir simulations for CO₂ storage in saline aquifers.

Introduction

As CO₂ emissions are constantly increasing from year to year, the problem of CO₂ sequestration gathers significant interest. One of the most discussed technologies is the CO₂ injection into deep aquifers, where CO₂ can be stored in various forms. Several trapping mechanisms exist in reservoirs, including structural and stratigraphic, residual gas, solubility, mineral, and hydrodynamic trapping (De Silva and Ranjith, 2012). However, the list of mechanisms of CO₂ transport in porous media includes more processes at various timescales, such as buoyancy, diffusion and dispersion (Nath et al., 2024). Comprehensive study of all these mechanisms is essential for accurate reservoir simulation, as prescribed by the international standard ISO 27914. Detailed special core analysis is of primary importance for the successful implementation of CO₂ storage in geological formations.

Relative permeability of CO₂/brine system is a critical input parameter for all hydrodynamic simulators (Burnside and Naylor, 2014). However, accurate measurement of relative permeability curves in gas/brine systems involves many uncertainties that must be minimized or at least quantified. Excellent reviews of experimental measurements of relative permeability in CO₂/brine systems are provided by (Burnside and Naylor, 2014; Müller, 2011).

A key challenge is that CO₂ typically exists in a supercritical state under reservoir conditions, characterized by increased solubility in brine, viscosity, and density. CO₂ solubility in brine complicates experimental procedures, as the phases should ideally be immiscible. High CO₂ concentrations also trigger competing mineral precipitation and dissolution reactions within the rock matrix (Bacci et al., 2011), altering porosity and permeability and thereby hindering accurate permeability and saturation monitoring. The increased CO₂ density, often close to that of brine, negatively affects saturation measurements using computed tomography (μ CT) or X-ray radiography — two state-of-the-art methods. When a third phase is present, measuring the saturation of all fluids under reservoir conditions becomes nearly impossible (Niu et al., 2010). Adding dopants such as potassium iodide (KI) to brine, especially low-salinity one, can improve contrast but alters brine properties by increasing viscosity and density, which impairs measurement quality. Mass balance methods are prone to high errors due to dry-out effects, dead-end volumes in coreflooding setup, and CO₂ expansion after exposure to ambient conditions, as demonstrated in our previous research (Penigin et al., 2024).

Another significant issue is the capillary end effect, caused by the strong affinity contrast of the rock surface for brine and CO₂. This effect creates a zone of preferential saturation at the core inlet, leading to incorrect permeability measurements due to increased flow resistance in this zone, as well as errors in saturation measurement using mass balance methods (Hinkley and Davis, 1986). Increasing the injection flow rate can mitigate this problem; however, this introduces a new challenge. As noted by (Jeong et al., 2021), the endpoint of the CO₂ relative permeability curve strongly depends on the injection flow rate: both relative CO₂ permeability and saturation increase with flow rate, with the most significant changes occurring at low flow rates. Thus, selecting the correct flow rate remains an open question.

Diffusion of CO₂ in water-saturated media is another important physicochemical factor that must be considered for long-term storage. While several studies address CO₂ diffusion in pure brine (Wang et al., 2025), the effects of porous media structure, pore channel tortuosity, and reservoir conditions are often overlooked.

This paper continues our research (Penigin et al., 2024), which examined CO₂ and brine interactions with siderite-containing sandstone rocks of moderate permeability. Here, we propose an approach for relative permeability measurement using two in situ saturation monitoring techniques, each capable of eliminating several uncertainties specific to CO₂/brine flow. In the second part of the paper, we quantify the diffusion properties of CO₂ in rock samples using experimental and numerical techniques.

Methods and Materials

Two experimental methods were used for saturation monitoring during relative permeability measurement: X-ray radiography and low-field NMR. For the X-ray radiography test, a PIK-SR radiographic complex, coupled with a PIK-OFP coreflooding setup (Geologika Ltd., Russia), was used with an X-ray transparent core holder. The X-ray system consists of the X-ray tube (160 kV) and the linear detector that captures 2D projections of the sample. The principal scheme of the setup is presented in Figure 1a. Core samples were loaded into the core holder, which was then connected to hydraulic lines for fluids and heaters. Two references were obtained during the experiment: a fully CO₂-saturated core immediately after loading and a fully brine-saturated core after CO₂ release and evacuation. Gas saturation during the experiment was calculated from projections according to Equation 1.

$$S_g = \frac{\ln(I_{exp}) - \ln(I_{brine})}{\ln(I_{gas}) - \ln(I_{brine})} \quad (\text{Eq.1})$$

where I_{brine} and I_{gas} are the intensities for brine and gas references, respectively. Each core sample projection was divided into a grid with a block size of 14×14 pixels (Fig 1b). Saturation was first calculated within each block and then averaged for the entire sample.

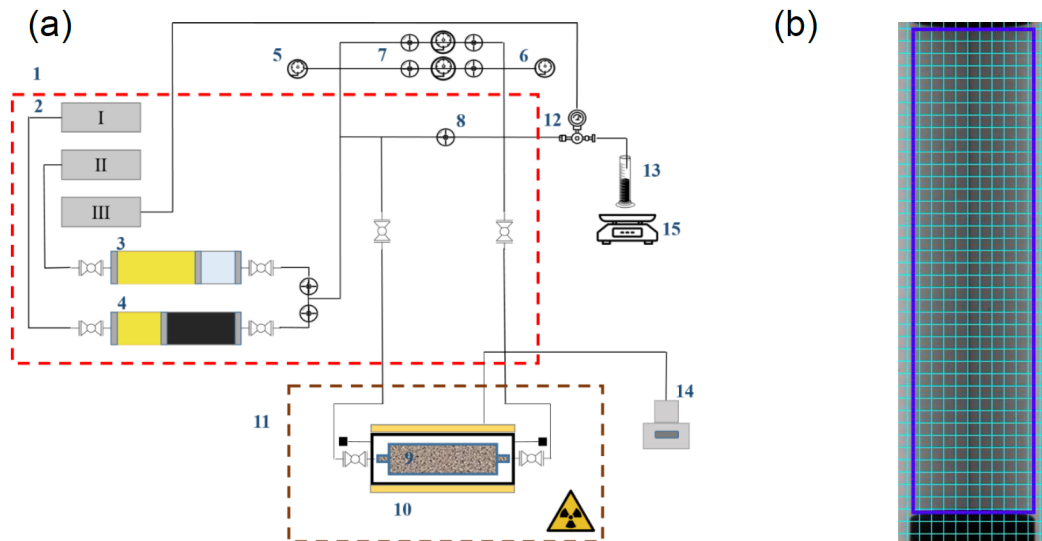


Figure 1—Experiment with X-Ray radiography: (a) scheme of the setup 1 – oven, 2 – plunger pumps, 3,4 – containers with fluids, 5,6 – pressure sensors, 7 – differential pressure sensor, 8 – bypass valve, 9 – core sample, 10 – coreholder, 11 – X-Ray module, 12 – backpressure regulator, 13 – fluids collect, 14 – confining pressure pump, 15 – balance, (b) grid of the 2D core projection.

Low-field NMR analysis was performed using a Geospec 2-53 NMR core analyzer (Oxford Instruments Inc). The 2 MHz NMR setup measures the signal resulting from resonance between the external magnetic field and hydrogen nuclei precession. As H_2O in brine contains hydrogen atoms, it can be detected with low-field NMR, whereas CO_2 remains transparent during two-phase flow. The core sample was packed into the Teflon-made sleeve and confined with perfluorinated oil (FC-40 by 3M). The sample was exposed to a constant magnetic field and number of sequences of radiofrequency pulses. Resonance signals were measured, and fluid volume was calculated in absolute units (cc). As water molecules in pores of different sizes have different tendencies to resonate in an external magnetic field, fluid distribution in pores of different sizes can be evaluated. Gradient coils allow for measurement of saturation profile along the selected core axis. T_2 relaxation curves are measured using the Carr-Purcell-Meiboom-Gill (CPMG) sequence (Abragam, 1961; Bloembergen et al., 1948; Straley, 1997). When measuring in a uniform field, the time-echo distance TE ($\text{TE}=2\tau$) was set equal to 0.1 ms. The minimum number of accumulations for T_2 was 64; for the saturation profile, it was 16. The NMR setup was coupled with a coreflooding system to maintain the reservoir conditions (Fig 2). The detailed workflow of HPHT NMR measurements is reported in (Mukhametdinova et al., 2025).

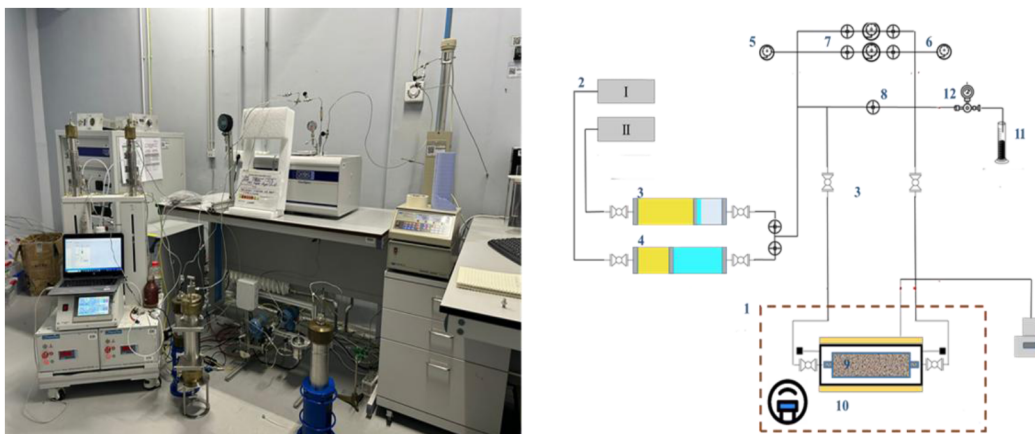


Figure 2—Experimental scheme of NMR core flooding setup: 1 – circuit of the NMR analyzer, 2 – plunger pumps, 3,4 – containers with fluids, 5,6 – pressure sensors, 7 – differential pressure sensor, 8 – bypass valve, 9 – core sample, 10 – coreholder, 11 – fluid collection, 12 – backpressure regulator, 13 – confining pressure pump.

Several core samples were used in this study. Their properties are summarized in Table 1. According to X-ray diffraction analysis, the samples consisted of quartz (75.1 wt.%), albite (10.4 wt.%), illite (7.6 wt.%), siderite (6.0 wt.%), and calcite (0.9 wt.%). The samples exhibited relatively high porosity and moderate permeability. In our previous paper (Penigin et al., 2024), it was shown that calcite is the most labile mineral, dissolving first upon contact with carbonated brine or CO₂/brine mixtures. Micro-computed tomography (μCT) analysis showed that the core samples were homogeneous, without fractures or inclusions present.

Table 1—List of core samples under study

Sample ID	Test	Porosity (%)	Permeability (mD)	Length (cm)	Diameter (cm)
1	X-Ray	18.67	10.38	3.05	2.98
2		18.82	10.34	3.04	2.98
3		18.06	9.28	3.08	2.98
4		18.10	9.14	3.00	2.98
5	NMR	18.66	12.34	2.67	2.42

CO₂ of 99.95% purity was used in both tests. The brine for the X-ray radiography test was prepared by dissolving 20 g/L of KI in distilled water, which is the minimum concentration required to ensure the necessary resolution for X-ray radiography with an uncertainty of $\pm 1.5\%$ in water saturation for system under study. For the NMR test, the brine was made by dissolving 20 g/L of KCl in distilled water, as K⁺ ions inhibit swelling and dispersion of the sensitive illite in the rock. In the relative permeability tests, both brines were carbonated to the maximum solubility of CO₂ under selected thermobaric conditions (1.45 mol/kg) to ensure that no mass transfer occurs between the phases.

Both relative permeability tests were carried out in stationary mode according to the following procedure:

1. Core holder assembly and loading into the coreflooding setup;
2. (For the X-ray radiography test) Saturation with CO₂ at reservoir pressure and temperature until constant volume was reached, followed by a reference scan ("CO₂");
3. Saturation of the core model with formation brine;
4. Injection of 20 pore volumes (PV) of carbonated brine to dissolve the most labile minerals ("Brine") until constant pressure drop;
5. Simultaneous injection of wet CO₂ and carbonated brine at ratios of 90/10, 75/25, 50/50, 25/75, 10/90, and 100/0. Each ratio was injected at two flowrates to quantify end effects as described by (Gupta

and Maloney, 2014). For the 100% gas ratio, flowrate was increased from lowest to highest; for other ratios, the order was reversed (highest to lowest) to establish equilibrium saturation faster;

6. In the X-ray radiography experiment, "reverse" (brine imbibition) relative permeability curves were obtained by simultaneous brine/CO₂ injection at the same ratios.

A special cell was created for diffusion tests; its scheme is shown in Figure 3a, and a photo is shown in Figure 3b. A core sample is sliced into a disk of 5 mm width and then inserted as a membrane between two parts of the cell. The cell is evacuated and then filled with nitrogen at atmospheric pressure. The cell is tested for leakage by sampling 50 µl of gas with a syringe from the left cell and checking its composition with gas chromatography. After 1 hour, pure CO₂ under atmospheric pressure starts circulating in the right cell, displacing nitrogen and establishing a 100% CO₂ boundary condition.

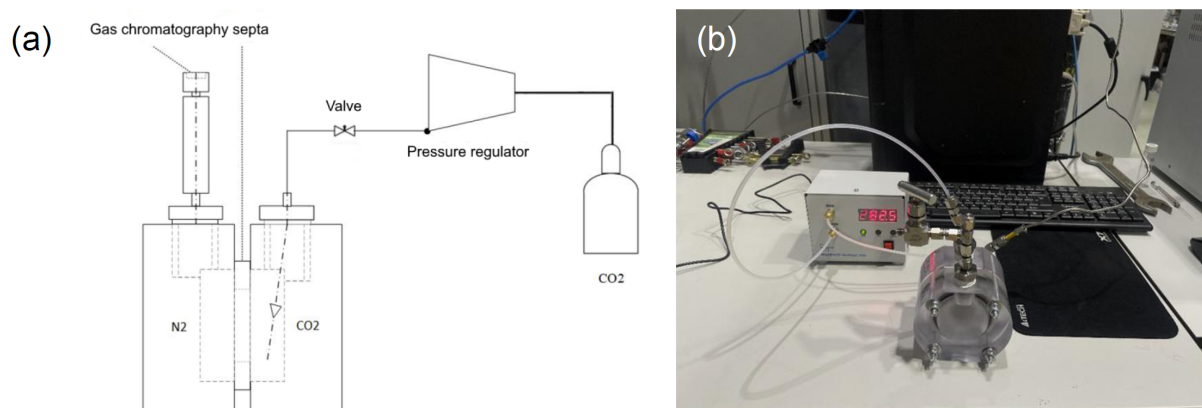


Figure 3—Cell for diffusion studies: (a) – scheme, (b) – experimental setup.

Gas probe was sampled from the left cell every five minutes and analyzed using gas chromatography. The diffusion coefficient was calculated by Fick's law (Equation 2), employing the data presented in Table 2.

$$\frac{\partial n}{\partial t} = DS \frac{\partial^2 n}{\partial x^2} \quad (\text{Equation 2})$$

Table 2—Input data for diffusion calculation

Parameter	Value
Core sample width, mm	6.182
Core sample diameter, mm	29.60
Cell volume (N ₂), cm ³	48
Duration, min	52
Temperature, °C	25
Absolute pressure, Pa	10.13*10 ⁴

Results

Relative permeability studies

Figure 4 presents the results of relative permeability measurements using X-ray radiography in the CO₂/brine system. The only parameter differentiating the two diagrams is the gas injection flowrate at the 100% gas stage. The experiment was conducted at 42°C and 22 MPa, ensuring that CO₂ was in a supercritical state. All permeability values were normalized by the initial permeability measured with carbonated brine.

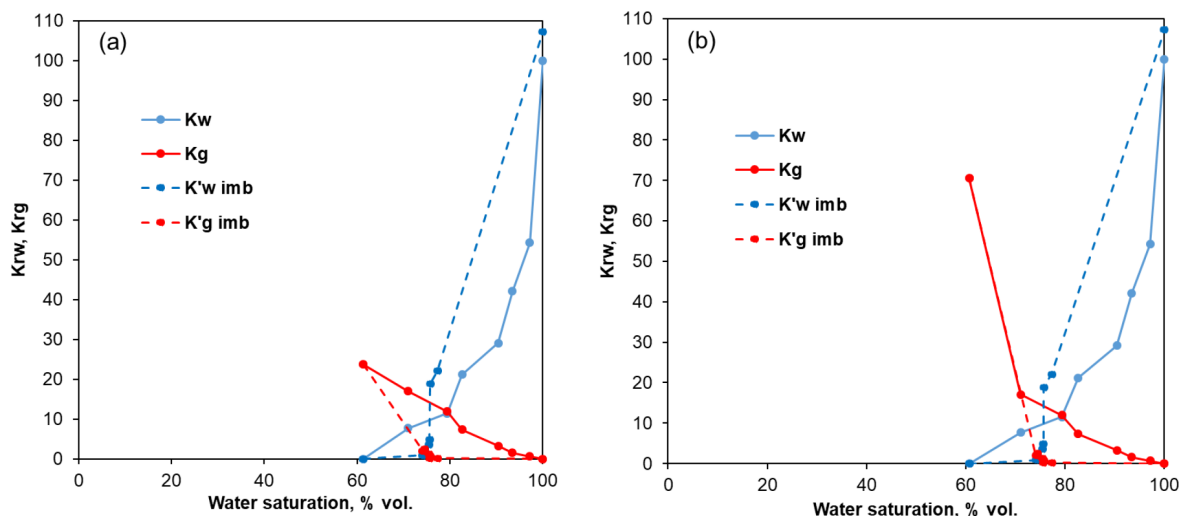


Figure 4—Relative permeability curves for CO₂/brine system in X-Ray radiography-assisted coreflooding: (a) 0.1 ml/min, (b) 1 ml/min gas injection flowrate at 100% gas injection stage.

As expected, increasing the flowrate raised the CO₂ endpoint relative permeability from 23.73% to nearly 70%. However, the water saturation changed minimally, decreasing only from 61.35% to 60.69%. This indicates that an order-of-magnitude increase in capillary forces caused the breakup of water globules in pore throats, which had previously hindered gas permeability. Although these globules occupy little pore space, their location obstructs gas flow. Once displaced, gas permeability improves.

Another important consideration is the selection of an appropriate flowrate to ensure unambiguous simulation results. In our view, the most reasonable flowrate corresponds to the linear velocity of the injected fluid in the reservoir, which is approximately 1 m/day. Therefore, the data in Figure 4a (0.1 ml/min) should be used as input for reservoir simulation. This choice also provides a conservative forecast for trapped CO₂ due to the highest residual water saturation observed at this flowrate.

Water imbibition after CO₂ drainage is notably restricted. The most likely cause is the Jamin effect, which inhibits water from penetrating channels occupied by gas—or, in this case, supercritical CO₂. According to (Burnside and Naylor, 2014), the trapped CO₂ saturation, defined as the CO₂ saturation at which its permeability approaches zero, is 24.3%, while the free CO₂ saturation is 14.2%. These conditions indicate favorable settings for CO₂ storage, as the risk of CO₂ migration due to water imbibition is minimal.

During this experiment, 2D projections (Figure 5a) revealed that approximately the first quarter of the core sample composite model was preferentially saturated with gas at all CO₂/brine ratios. This represents a clear end effect that should be excluded from analysis; therefore, saturation calculations were performed using the remaining three-quarters of the composite model length. Additionally, an intercept technique was applied to eliminate end effects (EE) in permeability measurements. The resulting relative permeability curves and its comparison with curves without end effects correction are presented in Figure 5b.

As can be seen, capillary end effects have a great influence on this system. First, the brine curve without eliminating end effects resembles the correct one; however, saturation values are shifted to the left, indicating lower water saturation. The gas curve has an entirely different shape. The most important reason for this effect is the endpoint: capillary end effects falsely increase gas saturation and gas permeability values. Therefore, the role of capillary end effects should be taken into account even when working with moderately permeable sandstone.

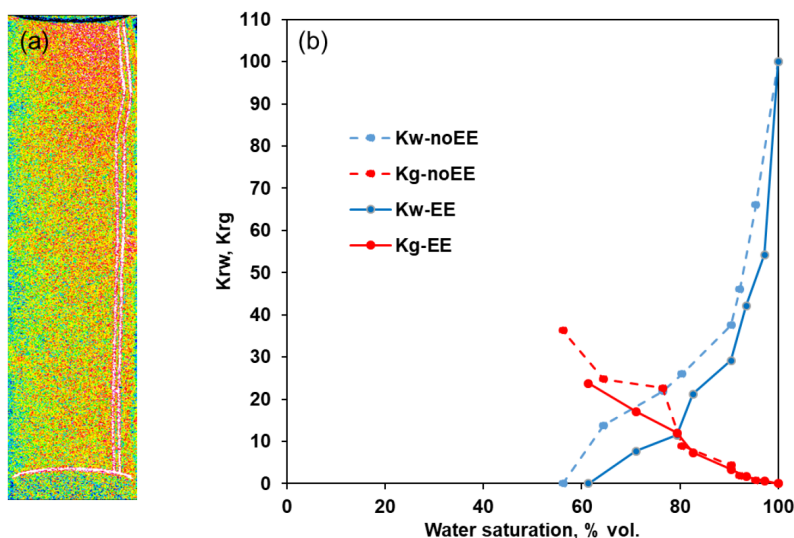


Figure 5—(a) Differential image of 2D projections at 10% brine / 90% gas stage and 100% brine stage, red area indicates higher gas saturation, flow occurs from top to bottom, (b) Relative permeability curves with (-EE) and without eliminating (no-EE) the influence of capillary end effects.

The second experiment was carried out with NMR in situ saturation monitoring. Relative permeability curves are shown in Figure 6a. The results are in good agreement with the curves obtained with X-ray radiography. Figure 6b presents the 1D distribution of hydrogen-containing fluid along the core axis. As can be seen, its concentration decreases at the inlet and outlet during flooding compared to the fully brine-saturated state; therefore, CO_2 concentration is higher in these regions. This fact indicates the capillary end effect observed earlier in the X-ray assisted experiment.

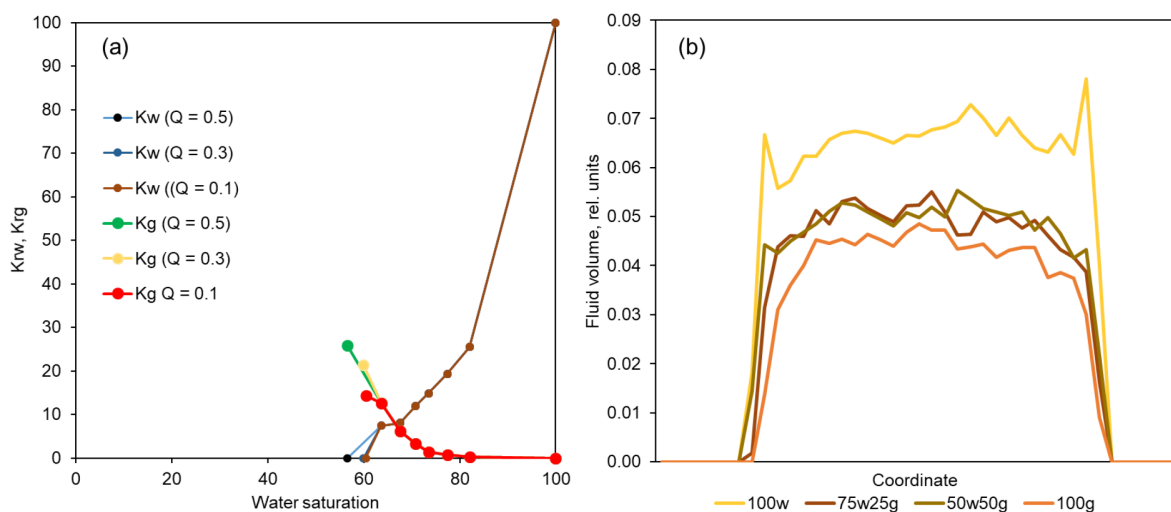


Figure 6—(a) Relative permeability curves for CO_2 /brine system at different flowrates, (b) 1D saturation profile along central axis of the core sample.

As can be seen in Figure 7a, after the start of 100% gas injection at a lower flowrate, water is completely removed from pores of large radius ($> 5 \mu\text{m}$). For example, if the flowrate is increased three times (from 0.1 to 0.3 ml/min), water is gradually displaced from pores of smaller radii (1–5 μm); however, small-radius pores ($< 0.7 \mu\text{m}$) are not influenced by gas injection. However, after such a small fluid removal, gas permeability increases from 0.33 to 0.6 mD, which is almost two-fold. Therefore, we can deduce that the primary reason for fluid flow resistance is concentrated in throats and small pores with radii of 1–5 μm , which are filled

with necks of water. After these necks are displaced because of the higher capillary number of CO₂ when flowrate is increased, phase permeability to CO₂ becomes greater.

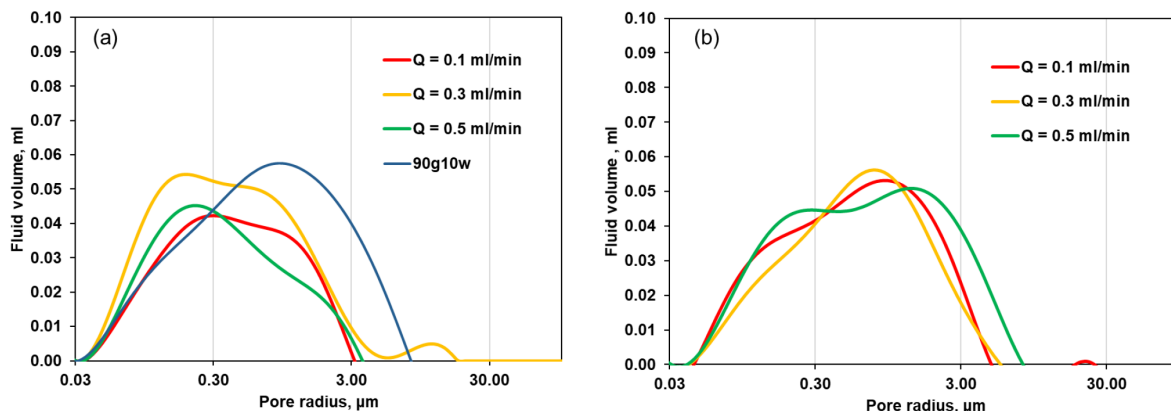


Figure 7—Fluid distribution in pores of different sizes at different flowrate
(a) 100% CO₂ injection at different flowrates, (b) 25% brine / 75% CO₂.

It was also noticed that this situation is specific of 100% gas injection, as, when two fluids are injected simultaneously at different flowrates, fluid distribution does not exhibit any trend to change (Figure 7b, for simultaneous injection of 75 brine / 25 gas).

Diffusivity studies

Figure 8a shows the change in CO₂ and N₂ concentrations in the left cell over time. As observed, CO₂ diffusion in the core sample is rapid, with approximately 50% of CO₂ diffusing within 15 minutes. The calculated diffusion coefficient for CO₂ was $2.83 \times 10^{-6} \text{ m}^2/\text{s}$, which aligns with the Chapman-Enskog approximation of $1.6 \times 10^{-5} \text{ m}^2/\text{s}$ if porous media tortuosity is taken into account.

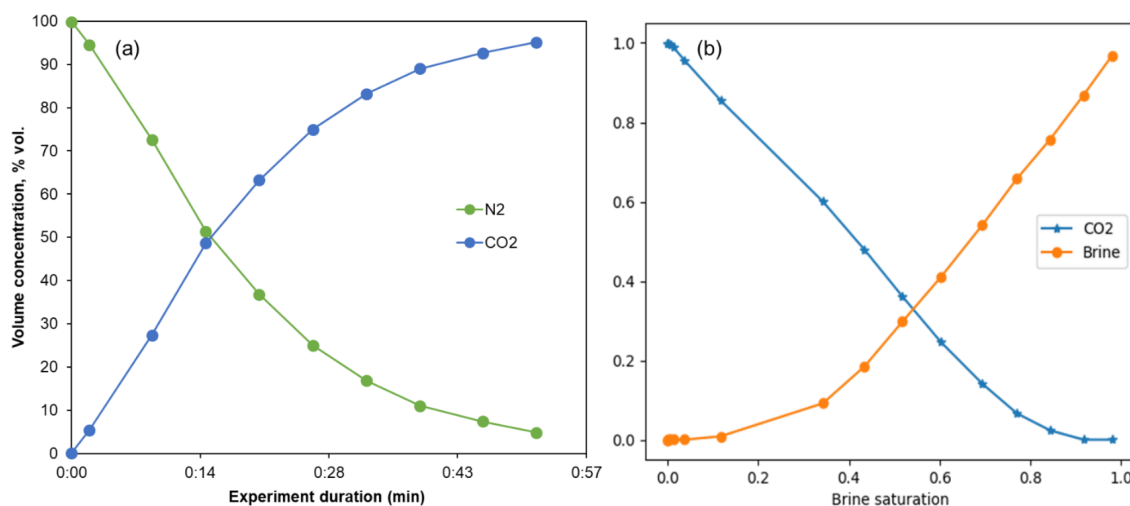


Figure 8—Diffusion study: (a) concentration of gas over time, (b) relative diffusivities curves for supercritical CO₂ – brine system.

However, this value diverges significantly from reservoir conditions for two primary reasons. First, the experiment studied gaseous CO₂, whereas reservoir conditions typically involve supercritical CO₂. Second, water saturation in porous media—common in reservoirs—hinders diffusion, complicating experimental measurements that require advanced techniques and prolonged timelines.

To address challenges in obtaining reliable diffusion data under water-saturated conditions, pore-network modeling (PNM) offers a practical solution. PNM represents porous media as interconnected pores (spherical) and channels (cylindrical) of varying radii. Fluid flow in this framework is governed by capillary forces, viscous flow, and diffusion. This method provides a computationally efficient approach to analyze mesoscale transport phenomena in porous systems.

In this work, we applied the OpenPNM Python package (Gostick et al., 2016) to evaluate the diffusion coefficient of supercritical CO₂ in porous media and its relative diffusivity in the presence of water. First, we created a block of pore system with a porosity of 15.6% and permeability of 12 mD, which closely matches the overall properties of the rock sample under study (Table 1). Then, using Fickian diffusion algorithms, we calculated the diffusion coefficient of gaseous and supercritical CO₂ (0.08 cP, 0.84 g/cm³) in this porous media. After that, using percolation algorithms and diffusive conductance models, we calculated the relative CO₂/water diffusivities in the porous media (Figure 8b).

The diffusivity value for gaseous CO₂ in the simulated porous media is 1.94×10^{-6} m²/s, which closely matches the experimental value. Therefore, the developed pore network model can be considered valid for further studies. The diffusion coefficient of supercritical CO₂ is 4.78×10^{-8} m²/s – two orders of magnitude lower. The relative diffusivity curves exhibit an almost X-shaped pattern. With knowledge of the effective single-phase diffusivities of water and supercritical CO₂, a diffusivity saturation-dependent function can be incorporated into reservoir simulations.

Conclusions

This manuscript investigates the flow dynamics of CO₂ and brine in sandstone rocks with moderate permeability, employing advanced laboratory techniques such as X-ray radiography and low-field NMR to analyze relative permeability, fluid distribution, and diffusion. The study highlights the impact of capillary end effects, flowrate variations, and pore-scale interactions on CO₂ sequestration efficiency, providing critical insights for reservoir simulation and long-term storage strategies.

- Higher injection flowrates significantly increase CO₂ endpoint relative permeability and only slightly reduce residual water saturation, emphasizing the need to align lab-scale flowrates with reservoir conditions for accurate simulation.
- Capillary end effects distort saturation measurements and relative permeability curves, necessitating their quantification and correction in core analysis to ensure reliable data.
- Water globules in pore throats (1–5 μm) are primary flow barriers; their displacement under higher flowrates enhances CO₂ permeability, while smaller pores (<0.7 μm) remain water-saturated.
- CO₂ diffusion in porous media is rapid in gaseous states but slows dramatically under supercritical conditions, with pore-network modeling offering a practical tool to predict diffusivity in reservoir scenarios.

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